



Rossmoyne Senior High School

Semester One Examination, 2019

Question/Answer booklet

MATHEMATICS METHODS UNIT 1

Section Two:

Calculator-assumed

SOLUTIONS

Student number: In figures

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In words

Your name

Time allowed for this section

Reading time before commencing work: ten minutes
Working time: one hundred minutes

Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet
Formula sheet (retained from Section One)

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: drawing instruments, templates, notes on two unfolded sheets of A4 paper, and up to three calculators approved for use in this examination

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	50	35
Section Two: Calculator-assumed	13	13	100	100	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answer to the specific question asked and to follow any instructions that are specified to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section Two: Calculator-assumed

65% (100 Marks)

This section has **thirteen (13)** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 100 minutes.

Question 9

(9 marks)

- (a) The points A and B have coordinates $(4, 5)$ and $(-5, 12)$ respectively. If A is the midpoint of B and C , determine the coordinates of C . (3 marks)

Solution
If $C(x, y)$ then $\frac{-5 + x}{2} = 4 \Rightarrow x = 13 \text{ and } \frac{12 + y}{2} = 5 \Rightarrow y = -2$ $C(13, -2)$
Specific behaviours
✓ writes equations for midpoints ✓ x-coordinate of C ✓ y-coordinate of C

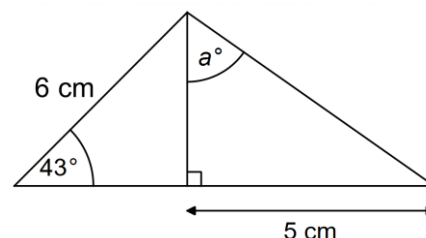
- (b) The points D and E have coordinates $(-3p, 2q)$ and $(4q, -p)$ respectively, where p and q are constants. Determine the value of p and the value of q if the midpoint of D and E is at $(-18, 7)$. (3 marks)

Solution
$\frac{-3p + 4q}{2} = -18 \text{ and } \frac{2q - p}{2} = 7$ Solve simultaneously CAS to get $p = 64, q = 39$
Specific behaviours
✓ equations for both coordinates of midpoint ✓ value of p ✓ value of q

- (c) Determine the value of a , correct to 2 decimal places.

(3 marks)

Solution
$\sin 43^\circ = \frac{x}{6} \rightarrow x = 6 \sin 43^\circ = 4.09$ $\therefore \tan a^\circ = \frac{5}{4.09} \rightarrow a = 50.70^\circ$
Specific behaviours
✓ calculates common side correctly ✓ calculates a correctly ✓ correct to 2 dp



Question 10

(9 marks)

In an experiment, the signal intensity, S , can be modelled by $S(x) = 7.7 - 6x + 2.1x^2 - 0.2x^3$, where x is the distance from the signal source in metres and $0 \leq x \leq 7$.

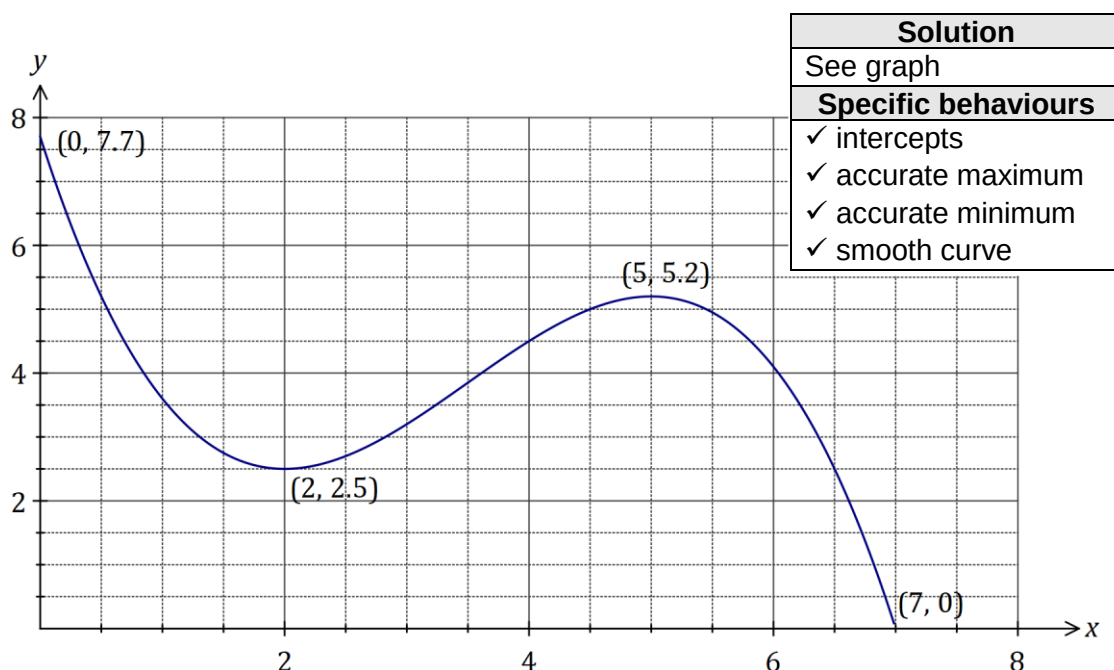
- (a) Determine S when $x = 6$.

(1 mark)

Solution
$S(6) = 4.1$
Specific behaviours
✓ correct value

- (b) Draw the graph of $y = S(x)$ on the axes below.

(4 marks)



- (c) Determine the equation of the straight line L that passes through the x -intercept and the y -intercept of the graph of $y = S(x)$.

(2 marks)

Solution
$(0, 7.7) \text{ \& } (7, 0) \Rightarrow m = -7.7 \div 7 = -1.1$ $y = -1.1x + 7.7$
Specific behaviours
✓ gradient
✓ y -intercept and equation

- (d) Determine the coordinates of the point of intersection of L with the graph of $y = S(x)$ where $x > 0$ and $y > 0$.

(1 mark)

Solution
$(3.5, 3.85)$
Specific behaviours
✓ correct coordinates

Question 11**(9 marks)**

A positive integer less than 12 is chosen at random.

The outcome sets for events A , B and C are $A = \{3, 4, 5, 6\}$, $B = \{1, 3, 5, 7\}$ and $C = \{2, 6, 10\}$.

(a) List the following sets:

(i) $A \cap B$.

(1 mark)

Solution
$A \cap B = \{3, 5\}$
Specific behaviours
✓ correct set

(ii) $A \cup B \cup C$.

(2 marks)

Solution
$A \cup B \cup C = \{1, 2, 3, 4, 5, 6, 7, 10\}$
Specific behaviours
✓ correct set
✓ uses curly braces to define set

(iii) $A \cap (B \cup C)$.

(2 mark)

Solution
$B \cup C = \{1, 2, 3, 5, 6, 7, 10\}$ $A \cap (B \cup C) = \{3, 5, 6\}$
Specific behaviours
✓ correct $B \cup C$
✓ correct answer

(b) Determine

(i) $n(B \cap C')$.

(1 mark)

Solution
4
Specific behaviours
✓ correct number

(ii) $P(B \cap C)$.

(1 mark)

Solution
$B \cap C = \{ \} \Rightarrow P(B \cap C) = 0$
Specific behaviours
✓ correct probability

(iii) $P(A' | (B \cup C))$.

(2 marks)

Solution
$P = \frac{4}{7}$
Specific behaviours
✓ denominator
✓ numerator

Question 12**(6 marks)**

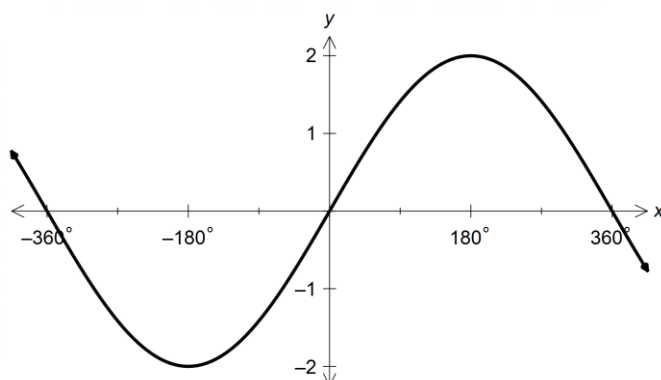
Determine the equations of each of the following trigonometric graphs.

(a)

(2 marks)

$$y = 2\sin \frac{x}{2}$$

✓ correct amp and recognises sine
✓ correct equation
(or $y=2\cos[1/2(x-180)]$ or others...)

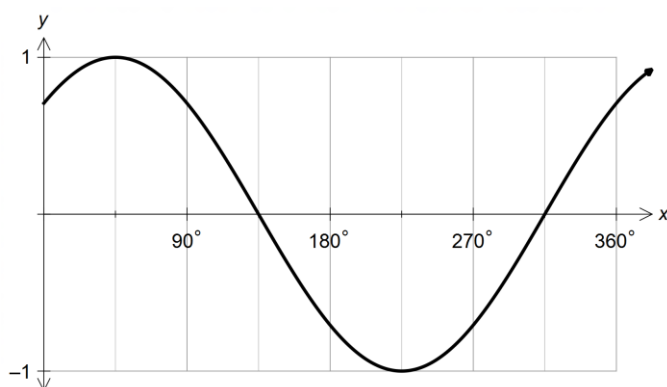


(b)

(2 marks)

$$\therefore y = \cos(x - 45^\circ)$$

✓ correct amp and recognises cos
✓ correct equation
(or $y=\sin(x+45)$ or)

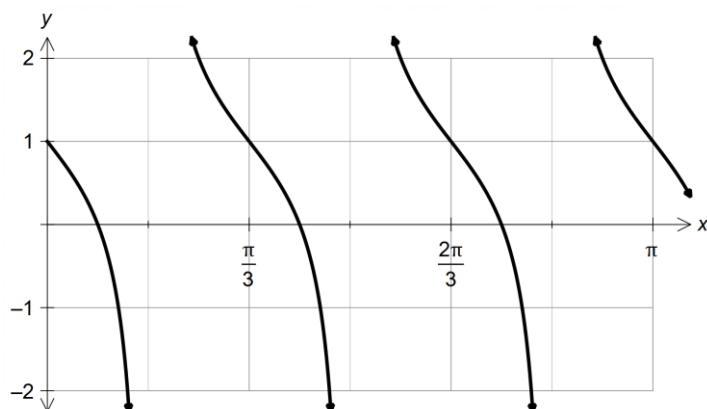


(c)

(2 marks)

$$y = -\tan 3x + 1$$

✓ recognises tan and -ve for reflection
✓ correct equation



Question 13**(7 marks)**

(a) Convert, giving an exact answer

(i) 16° to radians.

Solution
$\frac{4\pi}{45}$
Specific behaviours
✓ exact value

(1 mark)

(ii) 0.4 radians to degrees.

Solution
$\frac{72}{\pi}$
Specific behaviours
✓ exact value

(1 mark)

(b) Calculate, to the nearest degree, the acute angle between the line $y = 1.5x - 4$ and the x -axis. (2 marks)

Solution
$\theta_1 = \tan^{-1} 1.5 = 56$
Specific behaviours
✓ indicates use of $\tan^{-1} m$ ✓ correct angle with x -axis

(c) The sides adjacent to the right-angle in a right triangle have lengths 65 cm and 72 cm.

If the smallest angle in the triangle is α , determine an exact value for(i) $\tan \alpha$.

Solution
$\tan \alpha = \frac{65}{72}$
Specific behaviours
✓ correct ratio

(1 mark)

(ii) $\sin(90^\circ - \alpha)$.

Solution
$\sqrt{65^2 + 72^2} = 97$ $\sin(90^\circ - \alpha) = \frac{72}{97}$
Specific behaviours
✓ calculates length of hypotenuse ✓ correct answer

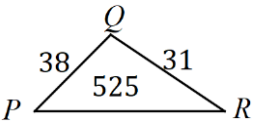
(2 marks)

Question 14**(7 marks)**

An **obtuse** angled triangle PQR has $p = 31$ cm, $r = 38$ cm and an area of 525 cm².

- (a) Sketch a triangle to show this information.

(1 mark)

Solution

Specific behaviours
✓ triangle with all information

- (b) Determine the size of $\angle Q$.

(2 marks)

Solution
$525 = \frac{1}{2}(38)(31) \sin Q$
$Q = 116.96 \approx 117^\circ$
Specific behaviours
✓ substitutes into area equation ✓ correct (obtuse) angle

- (c) Show that $q \approx 59$ cm.

(2 marks)

Solution
$q^2 = 38^2 + 31^2 - 2(38)(31) \cos 116.96$
$q = 58.93 \approx 59 \text{ cm}$
Specific behaviours
✓ uses appropriate equation that includes q ✓ substitutes correctly and solves to at least 1 dp

- (d) Show that $\angle P \approx 28^\circ$.

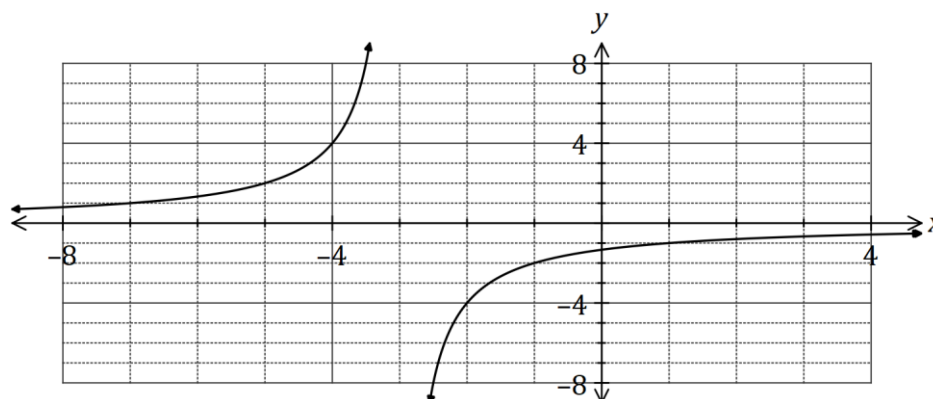
(2 marks)

Solution
$\frac{31}{\sin P} = \frac{58.93}{\sin 116.96}$
$\angle P = 27.96 \approx 28^\circ$
Specific behaviours
✓ uses appropriate equation that includes P ✓ substitutes correctly and solves to at least 1 dp

Question 15

(8 marks)

The graph of $y = f(x)$ is shown below where $f(x) = \frac{a}{b-x}$.



- (a) State the value of the constant a and the value of the constant b .

(3 marks)

Solution
$b = -3$
Using $(-4, 4) \Rightarrow 4 = a \div (-3 + 4) \Rightarrow a = 4$
Specific behaviours
✓ value of b ✓ uses point on curve ✓ value of a

- (b) The hyperbola shown above has two asymptotes. State their equations.

(2 marks)

Solution
$x = -3, \quad y = 0$
Specific behaviours
✓ vertical asymptote ✓ horizontal asymptote

- (c) Describe how to transform the graph of $y = f(x)$ to obtain the graph of $y = f(x) + 1$ and state the domain and range of the transformed function.

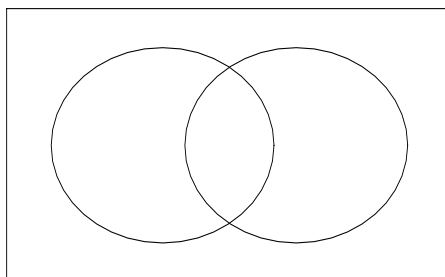
(3 marks)

Solution
Translate the graph 1 unit upwards.
Domain: $x \neq -3$
Range: $y \neq 1$
Specific behaviours
✓ transformation ✓ domain ✓ range

Question 16**(7 marks)**

An examination consisted of two papers, one of which was much harder than the other. 12% of candidates gained a distinction in the first paper and 4% gained a distinction in the second paper whilst 87% did not gain a distinction in either paper.

- (a) Represent the above information on the venn diagram below by placing a percentage in all regions. (2 marks)



- (b) Determine the probability that a randomly chosen candidate

- (i) gained a distinction in both papers. (1 marks)

Solution
$P(A \cup B) = 1 - 0.87 = 0.13$
$P(A \cap B) = 0.12 + 0.04 - 0.13 = 0.03$
Specific behaviours
✓ correct probability

- (ii) gained a distinction in one paper but not the other. (2 marks)

Solution
$P(A \cap \bar{B}) + P(\bar{A} \cap B) = (0.12 - 0.03) + (0.04 - 0.03) = 0.1$
Specific behaviours
✓ indicates correct method
✓ correct probability

- (iii) gained a distinction in the second paper given that they gained a distinction in the first. (2 mark)

Solution
$P(B A) = \frac{0.03}{0.12} = 0.25$
Specific behaviours
✓ numerator or denom correct
✓ correct probability

Question 17

(8 marks)

(a) The variables P and w are directly proportional and when $w = 30$, $P = 6$.

(i) Determine an equation for the relationship between P and w .

(2 marks)

Solution
$P = mw, \quad m = \frac{6}{30} = 0.2 \Rightarrow P = 0.2w$
Specific behaviours
✓ indicates use of line through origin ✓ correct relationship

(ii) State the value of P when $w = 10$.

(1 mark)

Solution
$P = 0.2(10) = 2$
Specific behaviours
✓ correct value

(b) The time, t minutes, that a car takes to travel 500 m at a constant speed of $s \text{ kmh}^{-1}$ is given by the formula $t = \frac{k}{s}$.

(i) Determine the value of the constant k , given that when $s = 30$, $t = 60$.

(1 mark)

Solution
$60 = \frac{k}{30} \Rightarrow k = 1800$
Specific behaviours
✓ correct value

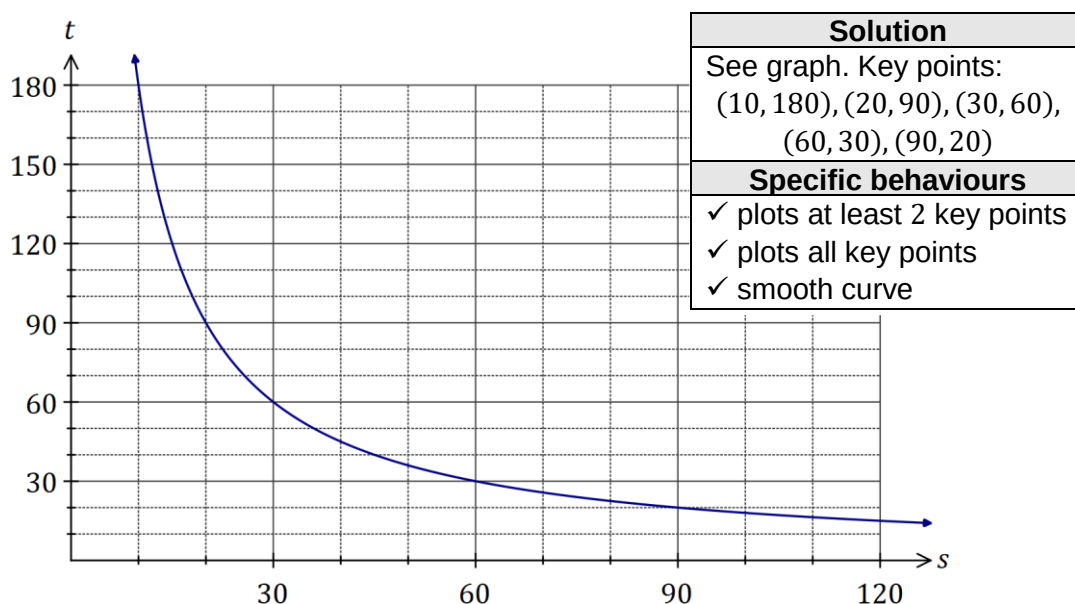
(ii) Determine the value of t when $s = 20$.

(1 mark)

Solution
$t = 1800 \div 20 = 90$
Specific behaviours
✓ correct value

(iii) On the axes below, draw a graph to show how s varies with t .

(3 marks)



Question 17

(8 marks)

Question 18

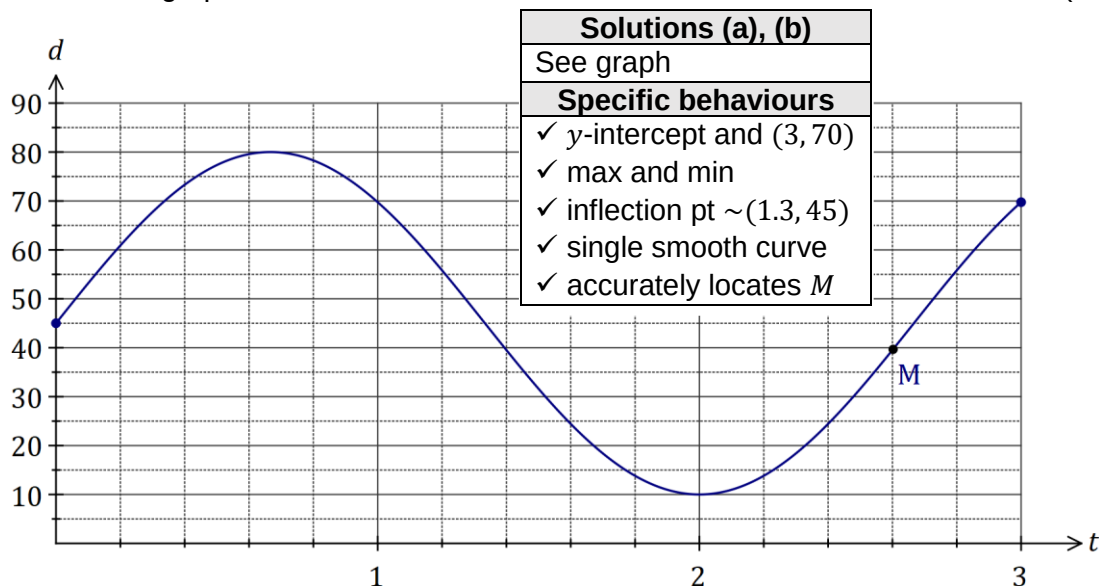
(8 marks)

A small weight, attached to the bottom of a spring, oscillated up and down. The distance, d cm, of the weight from the top of the spring after t seconds can be modelled by

$$d = 45 + 35 \sin\left(\frac{3\pi t}{4}\right)$$

- (a) Sketch the graph on the axes below for $0 \leq t \leq 3$.

(4 marks)



- (b) Mark on your graph point M , where the weight is 40 cm from the top of the spring and moving downwards.

(1 mark)

- (c) Determine

- (i) the maximum distance of the weight from the top of the spring.

(1 mark)

Solution
80 cm
Specific behaviours
✓ correct distance

- (ii) the time taken for the weight to first return to its initial position.

(1 mark)

Solution
$t = \frac{4}{3} = 1.\bar{3} \text{ s}$
Specific behaviours
✓ correct time

- (iii) the distance moved by the weight between $t = 1$ and $t = 2$.

(1 mark)

Solution
$d(1) - d(2) = 69.75 - 10$ $= 59.75 \text{ cm}$
Specific behaviours
✓ correct distance

Question 19
(8 marks)

- (a) The equation of the axis of symmetry for the graph of $y = 2x^2 + 8x + 5$ is $x = m$. Determine the value of m , using a method that does not refer to the graph of the parabola. (2 marks)

Solution	
$x = -\frac{8}{2 \times 2} = -2$ $m = -2$	$y = 2(x + 2)^2 + c$ $\therefore m = -2$
Specific behaviours	
✓ uses $x = -b \div (2a)$ or partially completes the square ✓ value of m	

- (b) A parabola with equation $y = ax^2 + bx + c$ has a turning point at $(4, -5)$ and passes through the point $(2, -17)$. Determine the value of a , the value of b and the value of c . (3 marks)

Solution
$y = a(x - 4)^2 - 5$ $-17 = a(2 - 4)^2 - 5 \Rightarrow a = -3$ $y = -3(x - 4)^2 - 5$ $= -3x^2 + 24x - 53$ $a = -3, \quad b = 24, \quad c = -53$
Specific behaviours
✓ correctly writes in turning point form ✓ solves for a using point ✓ expands and states all values

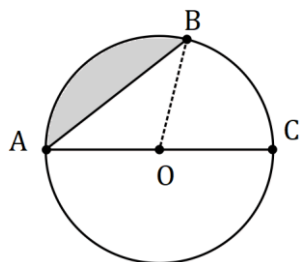
- (c) Determine the value of the discriminant for the quadratic equation $4x^2 - 28x + 47 = 0$ and use it to explain how many solutions the equation $(x + 3)(4x^2 - 28x + 47) = 0$ will have. (3 marks)

Solution
$d = (-28)^2 - 4(4)(47) = 32$ When $d > 0$, quadratic will have two solutions. Hence equation will have three solutions - one from the linear factor and two from the quadratic factor.
Specific behaviours
✓ value of discriminant ✓ uses discriminant to say quadratic will have two solutions ✓ explains why equation has three solutions

Question 20

(8 marks)

- (a) The circle shown has centre O and diameter AC of length 80 cm. Determine the shaded area given that $3 \times \angle AOB = 5 \times \angle BOC$. (4 marks)



Solution
$\angle BOC = \frac{3}{5}\angle AOB$ $\angle AOB + \frac{3}{5}\angle AOB = \pi \Rightarrow \angle AOB = \frac{5\pi}{8} = 112.5^\circ$ $A = \frac{1}{2}(40)^2 \left(\frac{5\pi}{8} - \sin \frac{5\pi}{8} \right)$ $= 832 \text{ cm}^2$
Specific behaviours
✓ equation using angles ✓ correct angle for segment ✓ substitutes correctly into formula ✓ correct area

- (b) A sector of a circle has a perimeter of 84 cm and an area of 360 cm^2 . Determine the radius of the circle. (4 marks)

Solution
$2r + r\theta = 84$ $\frac{1}{2}r^2\theta = 360$ Solving simultaneously gives $r = 12, \theta = 5 \text{ or } r = 30, \theta = \frac{4}{5}$ Hence $r = 12$ or $r = 30$ cm
Specific behaviours
✓ equation for perimeter ✓ equation for area ✓ solution of equations ✓ states both values of r

Question 21

(6 marks)

Let $a = \sin 50^\circ$ and $b = \cos 100^\circ$.

Give your answers to the following in terms of a and/or b .

(a) Write down an expression for

(i) $\sin 130^\circ$.

(1 mark)

Solution
$\sin 130^\circ = \sin 50^\circ = a$
Specific behaviours
✓ correct expression

(ii) $\cos 80^\circ$.

(1 mark)

Solution
$\cos 80^\circ = -\cos 100^\circ = -b$
Specific behaviours
✓ correct expression

(b) Determine an expression for $\cos 130^\circ$.

(3 marks)

Solution
$\cos^2 130^\circ + \sin^2 130^\circ = 1$ $\cos^2 130^\circ = 1 - a^2$ $\cos 130^\circ = \pm\sqrt{1 - a^2}$ but $\cos 130^\circ < 0 \Rightarrow \cos 130^\circ = -\sqrt{1 - a^2}$ or $\cos(100+30)=\cos 100\cos 30-\sin 100\sin 30$ $= \frac{\sqrt{3}b}{2} - a\sqrt{\frac{(b+1)}{2}}$
Specific behaviours
✓ indicates use of $\sin^2 \theta + \cos^2 \theta = 1$ or angle sum ✓ expression for $\cos^2 130^\circ$ or angle sum ✓ correct expression

(c) Determine an expression for $\tan 130^\circ$.

(1 mark)

Solution
$\tan 130^\circ = \frac{\sin 130^\circ}{\cos 130^\circ} = -\frac{a}{\sqrt{1 - a^2}}$
Specific behaviours
✓ correct expression

Supplementary page

Question number: _____

Supplementary page

Question number: _____

Supplementary page

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